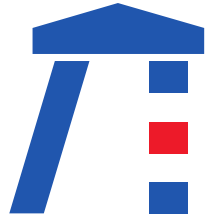
# Mesh Network

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**Seminar: Verteilte und vernetzte Systeme**



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# Outline

1. **Why Mesh Networks?**
   - Existing System
   - iMesh
2. **Architectural Overview**
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   - Mesh Formation
   - Internetworking
   - Path Selection Mechanism
   - Medium Access Control
   - Other Features
3. **OLPC-Mesh**
4. **SMesh**

# Why Mesh Networks?

- Shift from Ethernet cables to wireless communication
- In 2003, the 802.11 working group defined the idea of Wireless Distribution System (WDS) as a mechanism for wireless communication using a four address frame format. But did not describe how use it
- The benefits were **flexible**, **self-forming**, **self-healing** network

# Existing Systems

- Infrastructure mode, Ad-hoc, Mixed mode
- Uses layer 3 (IP based) routing

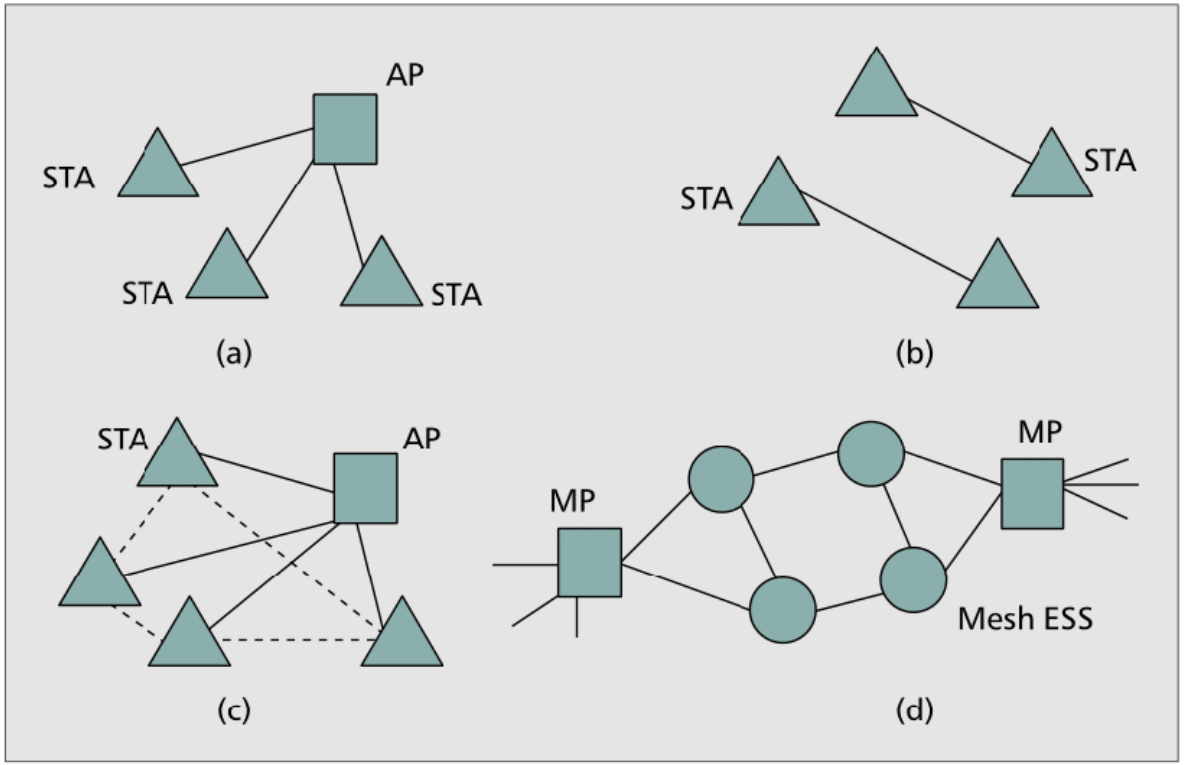



Figure: comparing Mesh with existing systems; a) Infrastructure mode, b) Ad-hoc, c) Mixed mode, d) Mesh Network

# Challenges

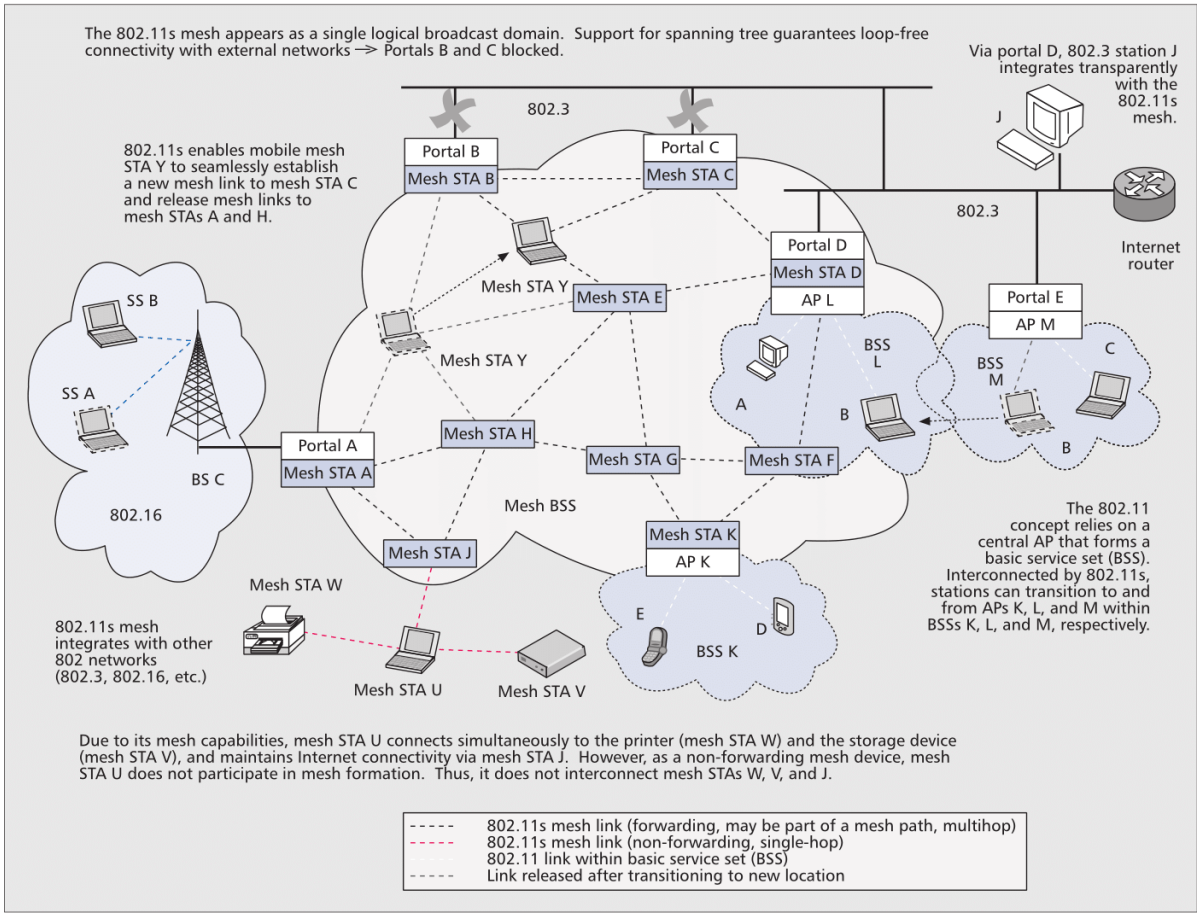
- Major challenges were :
  - Routing mechanism base on layer 2 (MAC)
  - Different Link metrics approaches
  - Routing protocols depending on the needs
  - QoS mechanisms depending on demand (eg; Real time)
  - Security and power efficiency

# iMesh

- Earlier attempt for creating mesh on IEEE 802.11.
- With fundamental design goal to pursue client transparency
- Bridging, layer-2 alternative to routing has issues, so using layer-3 handoff
- Link layer handoff achieved by probing
- Network layer handoff achieved Transparent Mobile IP (TMIP) with help of Mobile Location Register (MLR)
- APs will have a IP to MAC address mapping table for its "nodes"
- Optimizied Link State Routing (OLSR).

# Architectural Overview

- Nodes in Mesh are categorized into Station (STA), Mesh Station (Mesh STA), Mesh Access Point (Mesh AP), Portal



# Frame structure

- Extension to the existing 802.11 frame structure
- 802.11 categorizes frames as data, control or management
- 802.11s extends data (Mesh Data Frame) and management frame (Mesh-specific Management frame) by an additional Mesh control field
- Mesh control field contains Time-to-live (TTL), Mesh sequence number, a Mesh flags field and possibly a Mesh address extension field

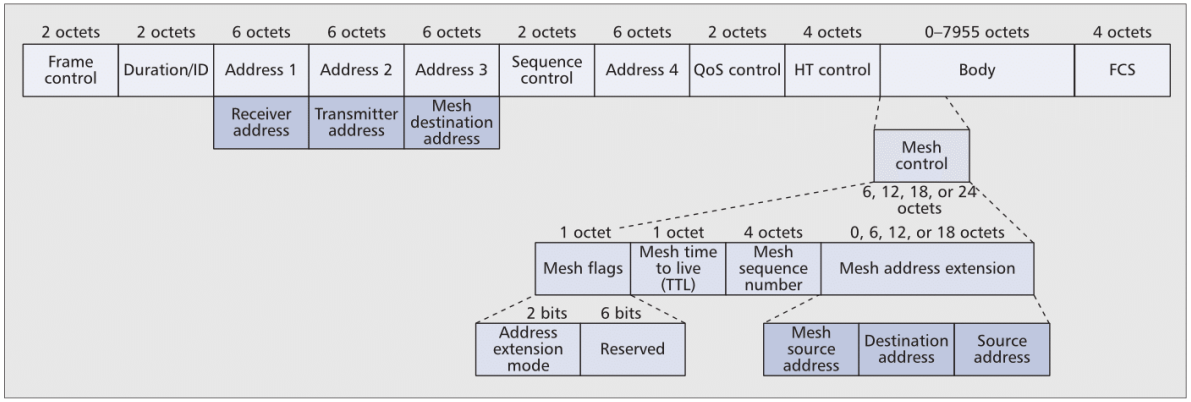



Figure: 802.11s frame

# Frame structure

- Supports four or six MAC address frame formats
- Four-address frame format :
  - Source Address (SA)
  - Destination Address (DA)
  - Transmitter Address (TA)
  - Receiver Address (RA)
- Apart from four address, six-address frame format has:
  - Mesh SA
  - Mesh DA

# Mesh Formation

- Similar to Beacon frame, Mesh uses Mesh Identifier (Mesh ID)
- Profile consists of 1) Mesh ID, 2) Path Selection Protocol, 3) Path Selection Metric
- Mesh STA different profile but only one at a give moment
- Default: Hybrid Wireless Mesh Protocol (HWMP)
- Similar neighbor discovery mechanism as 802.11
- Mesh Peer Link Management Protocol to open a Mesh peer link

# Mesh Formation

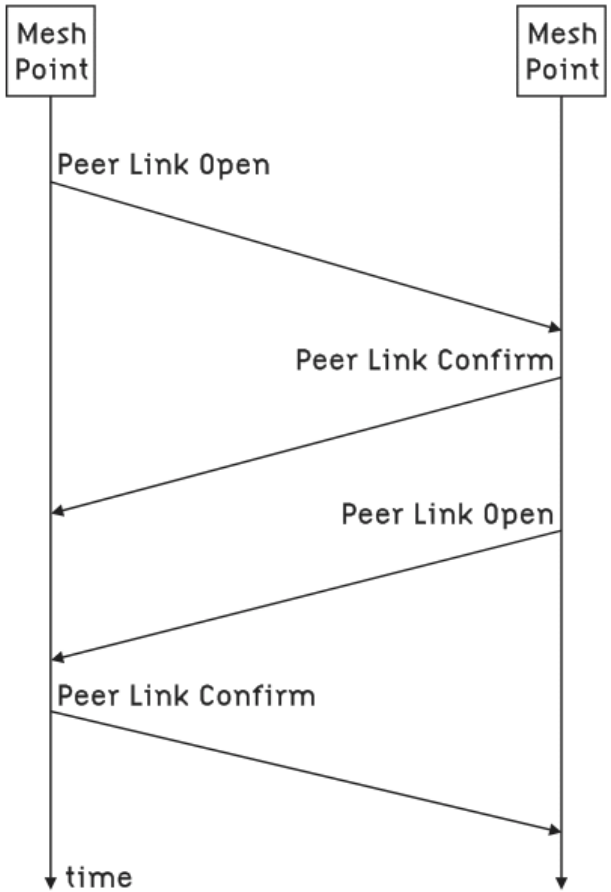



Figure: Establishment of a peer link in 802.11s

# Internetworking With Mesh

- Using Portal to interconnect Mesh clouds to other LAN
- Portal act as gateway to different layer-three subnets
- Portal uses Portal Announcement (PANN )frame spread its role
- Portal also connects Mesh networks with running different protocols

# Internetworking With Mesh

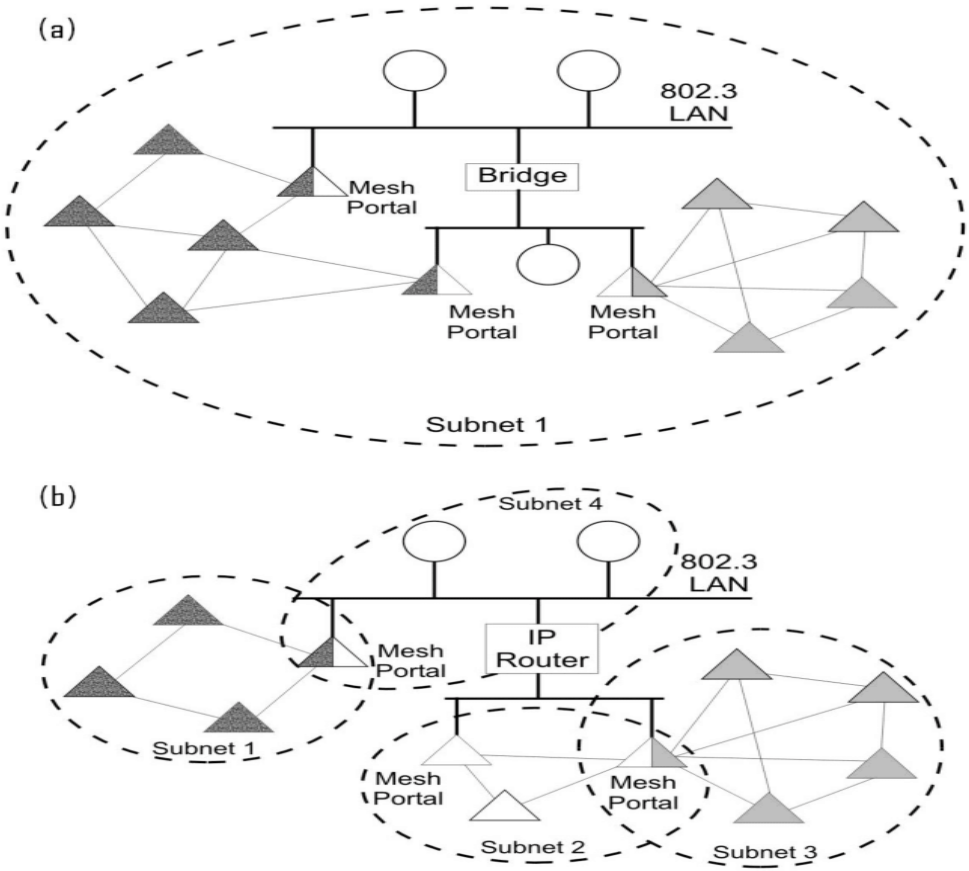



Figure: Internetworking; a) layer 2 bridging b) Layer 3 internetworking

# Path Selection Mechanism

- Default: HWMP (Hybrid Protocol) and the Airtime Link Metric
- Configuration element is transported by beacon frames, Peer Link Open Frames and Peer Link Confirm frames

# HWMP

- HWMP inspired from Ad Hoc On Demand Vector (AODV) protocol and its extensions.
- Two modes;
  - 1) on-demand reactive mode, appropriate for establish path between Mesh STAs in a peer-to-peer basis and
  - 2) tree-based proactive mode, better for tree based topology. Further subdivided into:
    - Proactive PREQ mechanism
    - Proactive RANN mechanism

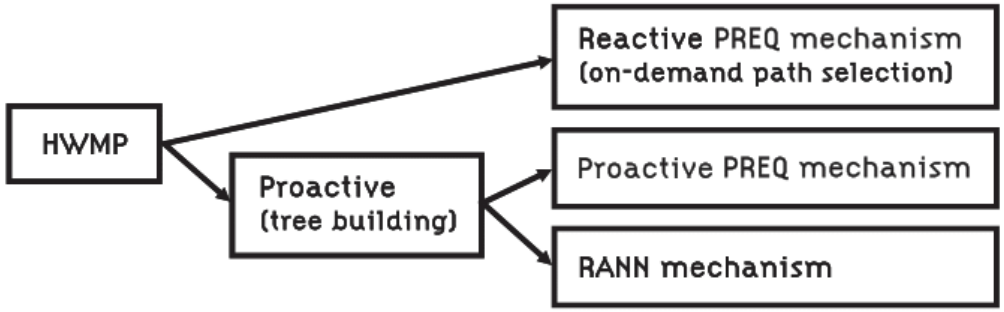



Figure: HWMP path selection mechanism

# HWMP Management frames

- Four elements:
  - Path Request (PREQ)
  - Path Reply (PREP)
  - Path Error (PERR)
  - Root Announcement (RANN)

# On Demand path discovery

- S sends PREQ frames, depending DO (Destination Only) flag, intermediate node reply PREP frame.
- Reply and Forward (RF) used to control the behavior of intermediate nodes
- Both PREQ AND PREP carry metric value for path selection

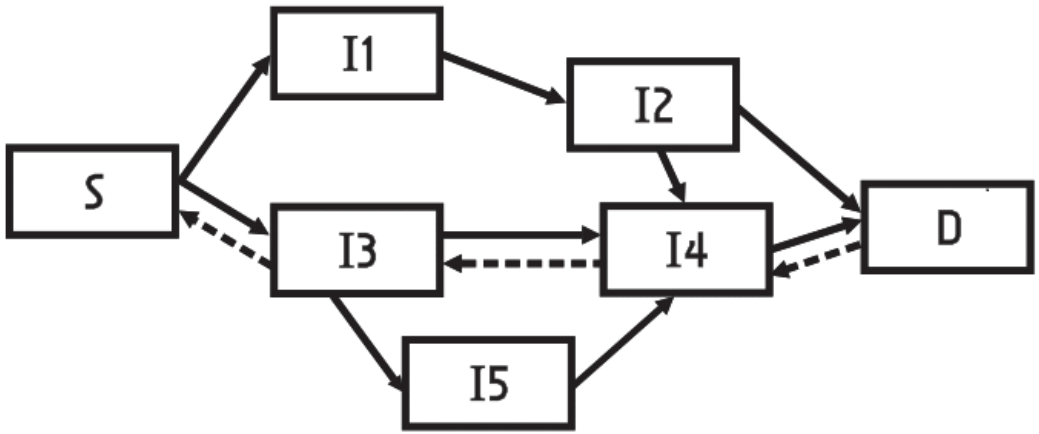



Figure: On demand path discovery S $\rightarrow$ D

# Proactive mechanisms

- Proactive PREQ mechanism:
  - Root node broadcast PREQ frame
  - RF and DO set to 1
  - Proactive PREP depends on root settings
- Proactive RANN mechanism
  - Root node broadcast RANN frame
  - Interest nodes in path formation reply with PREQ (same rules)
  - Root reply PREP

# Airtime Link Metric

- depends on :
  - Amount of time consumed to transmit a test frame
  - and its value takes into account the bit rate at which the frame can be transmitted,
  - the overhead posed by the PHY implementation in use and also the probability of retransmission, which relates to link error state
- $$c_a = [O + \frac{B_t}{r}]\frac{1}{1 - e_f}$$
  - O is a constant overhead latency
  - $B_t$ is the test frame size
  - r is the data rate Mb/s
  - $e_f$ test frame error rate

# Medium Access Control in 802.11s

- For medium access, uses Mesh Coordination Function (MCF)
- MCF has two parts; Mandatory and optional
  - For the mandatory part, MCF relies on Enhanced Distributed Channel Access (EDCA)
  - Limited Transmission Opportunity (TXOP).
  - To enhance QoS, MCF describes optional medium access protocol called Mesh Coordinated Channel Access (MCCA)
  - Mesh STA can reserve TXOP for future called MCCA Opportunities (MCCAOPs)

# Other Features

- Synchronization
  - Optional.
  - At present, time reference is send along with beacon frames
  - Similar to 802.11
- Power Management
  - Depending on the routing protocol
  - Normally power saving Mesh nodes be light-sleep mode or deep-sleep mode

# Other Features

- Congestion Control
  - uses management frames
- Security
  - Similar to 802.11
  - DoS, Signal Jamming, Wormhole attack etc...

# OLPC-Mesh

- OLPC's XO was the first device to implement a mesh network based on the IEEE 802.11s
- has its own characteristics and diverges from IEEE 802.11s draft
- Path asymmetry for HWMP
- Different link metric calculation for link cost

# OLPC-Mesh

- Path discovery
  - XO will send a set of PREQs, four frames at different transmission rate (54, 36, 11 and 1 Mbps with cost 13, 28, 42 and 64 respectively) called PREQ cluster
  - Intermediate nodes re-broadcast frames received from source (S) with a certain delay (rreqdelay), Network Wide Broadcast (NWB) and after calculating the cost depending on data rate the frame received
  - No DO and RF flags in path discovery frames
  - Once the frame received at destination (D), similar mechanism as IEEE 802.11s proposal but new path discovery for D $\rightarrow$ S

# OLPC-Mesh

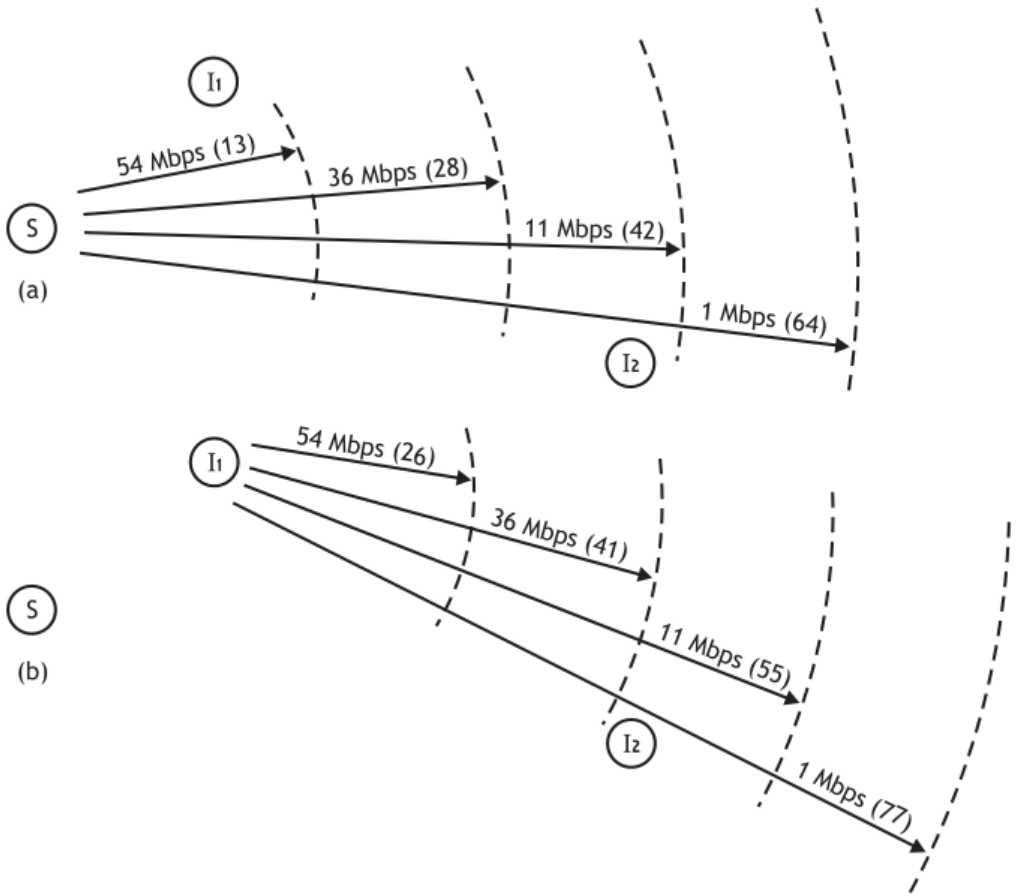



Figure: OLPC path discovery S $\rightarrow$ D

# OLPC-Mesh

- Mainly for laptops with limited processing power
- OLPC has no knowledge about link as being established
- Several standard were not applied

# SMesh

- SMesh or Seamless Mesh developed by the Distributed System and Networks Lab at Johns Hopkins University
- Operates in regular 802.11 IBSS mode and provides peer-to-peer connectivity, Internet connectivity, and fast handoff to mobile clients across the mesh
- Communication infrastructure relies on Spines message system
- Mobile clients gets an illusion being stationary by providing same connectivity information to clients through DHCP
- Focused mainly for real-time applications like VoIP, Video conferencing

# SMesh Architecture

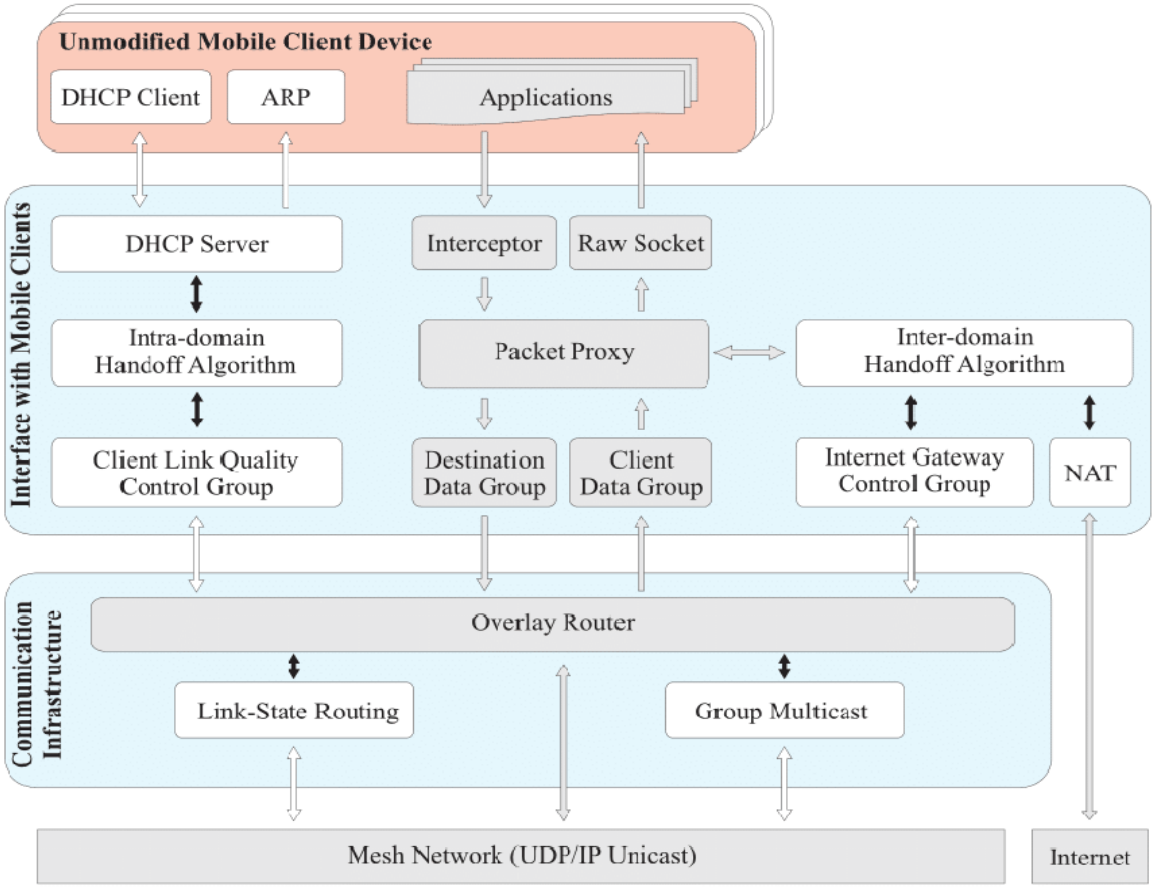
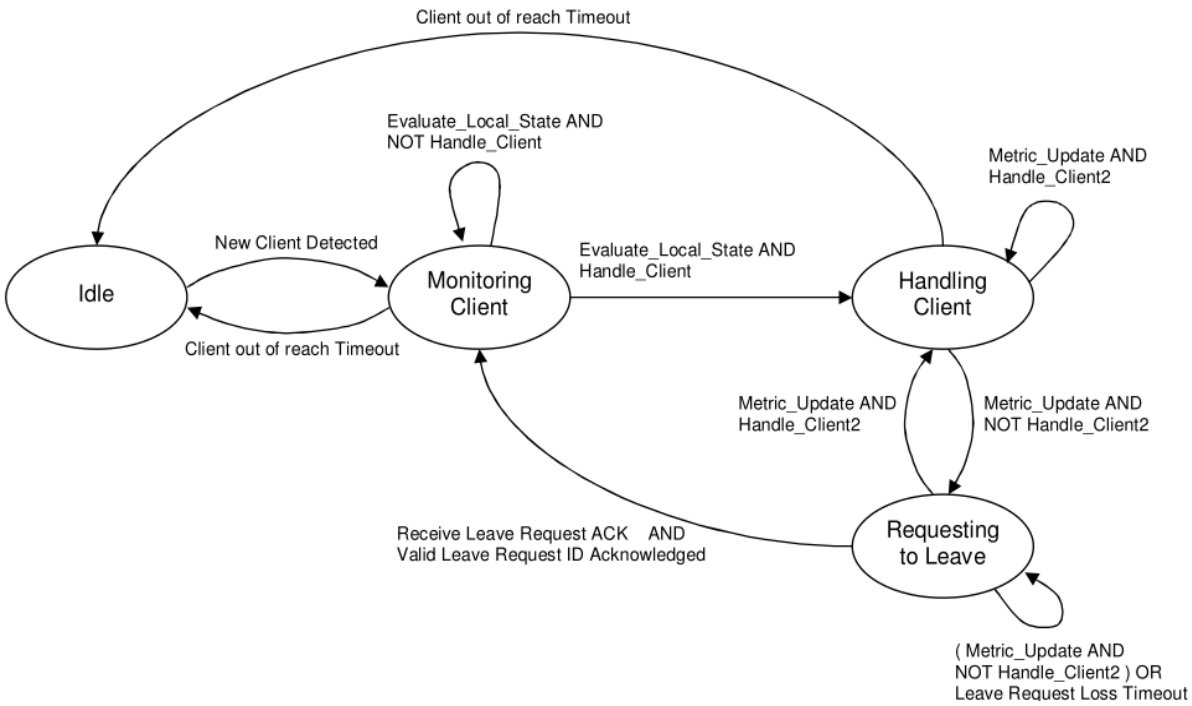



Figure: Architecture of SMesh

# Fast Intra-Domain Handoff protocol

- Mobile Client Monitoring
  - Instead of client "decide" when the handoff should take place, protocol decide when to change
  - using DHCP and ARP protocol, force to use a single IP for a client
  - uses special metric for calculating the quality of link depending on quality of DHCP requests or ARP response received by the client
- Intra-domain Mobility Management
  - Uses Client Data Group (CDG), Client Control Group (CCG)
  - Client will join to a set of two unique multi-cast group (CDG + CCG)
  - Client handoff achieved using a virtual IP in DHCP broadcast.
  - Performance of handoff depends on decision making.

# SMesh Architecture




Conditions

| | |
|---|---|
| Handle_Client: | My_Metric > Highest_Metric(Data Group) * Threshold AND<br>My_Rank(Nodes in MonitoringClient state) <= Maximum_Concurrent_Joins (in our case 2) |
| Handle_Client2: | My_Rank(Nodes in HandlingClient or RequestingToLeave state) == 1 |
| My_Rank(list): | sort list in decreasing order of their metric value, then by IP address to break ties, and return index of local node |

Figure: State machine of handling mobile clients

# Fast Inter-Domain Handoff protocol

- Communication between mobile client and Internet, resides on client's private IP and NAT of the Internet gateway
- Movement of client from one gateway to other gateway region need to notified
- Need a solution to support real-time traffic
- TCP Connection Handoff
  - "Slowly move the connection"
  - If client changes its gateway with a existing TCP connection, redirect the packet through old gateway, and if it starts a new connection use the new gateway.
  - By multi-casting to Internet Gateways Multicast Group (IGMG), also performance depends on the data rate at Internet gateway

# SMesh Architecture

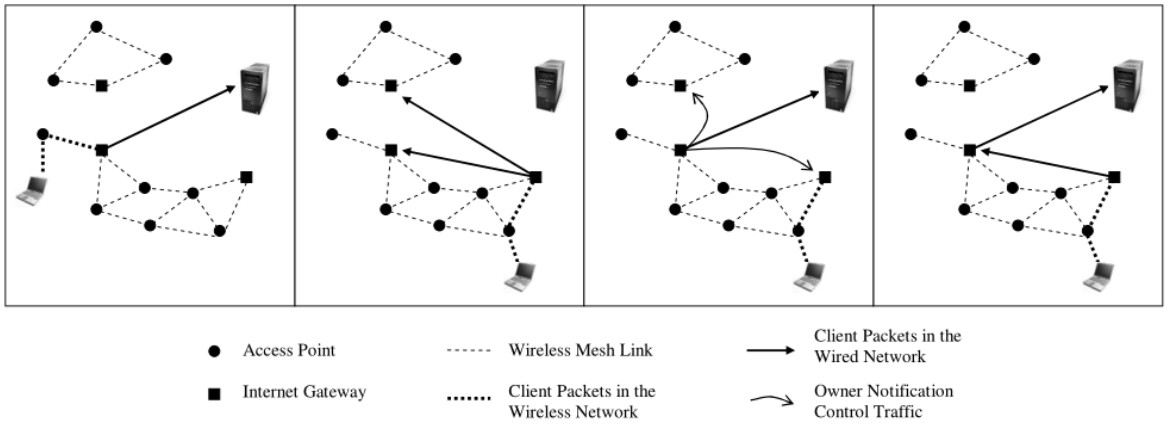



Figure: TCP forward handoff: (a) Connection establishment (b) Handoff Phase 1 (c) Handoff Phase 2 (d) Handoff completed

- UDP Connection Handoff
  - Classify UDP traffic on a port basis as connection-less and connection-oriented, default as connection-oriented, else forward directly
  - Upon receiving a connection-oriented UDP, relays packet to destination and IGMP multicast group.
  - By multi-casting to Internet Gateways Multicast Group (IGMG), also performance depends on the data rate at Internet gateway. Proper decision is made afterwards

# SMesh

- Overall
  - a good attempt for real-time traffic
  - issues like maintaining duplicate vs handoff performance, security issues, crashing of nodes, Management overhead....


Thank you!